

1. Computer vision ~~uses~~ helps computers to understand the photos and videos and extract meaningful information

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a) objectives

- Image understanding
- object detection
- Scene interpretation

b) Two-Real world Applications

- Facial recognition
- self-driving cars.

2. It can extract features, recognize objects and make decisions from visual data

a) Feature extraction

Feature extraction is the process of identifying important characteristics from image. (edges, corners etc..)

b) It helps system identifying the objects and classify it accurately

3) There are three levels of image processing
low level, mid level, High level

a) low-level image processing

low-level image processing improves image quality by removing noise and enhancing images

b) Mid-level: segmentation and feature extraction

High-level: object recognition and scene understanding

4) A digital image is represented as a grid of pixels

a) Pixel: A pixel is the smallest element of a digital image

b) Higher resolution gives better quality and more details

5) Image sensing and acquisition converts real world scene into a digital image by capturing light from scene

a) Image sensor

An Image sensor converts light into electrical signal

b) Two Image Acquisition Devices

- Digital camera
- scanner.

6) Sampling and quantization help convert an analog image into a digital image

a) Sampling:

Sampling is the process of dividing image into pixels

b) Quantization in image

Quantization assigns intensity values to the pixels.

It explains hows light from object forms an image on a sensor. It Explains hows camera capture the image of real world.

a) Perspective Projection

A method that projects a 3D scene onto a 2D image plane.

b) illumination in image formation

illumination provides the light needed to capture visible details.

8) Image Processing

Image processing involves manipulating images to improve or analyze them

b) Two differences

Image processing

1. enhances images.
2. focuses on image quality

computer vision

1. understands image.
2. Focuses on extracting information.

9) a) It used to detect diseases from medical images such as x-rays and MRI scans

b) It helps vehicles identify roads, traffic signs etc..

10) They provide the visual data needed for analysis and recognition.

a) Every digital image is made up of pixels

b) Higher sampling and resolution provide more details and improve analysis accuracy.